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Metadata S1

18-year plant reproductive phenology dataset from Lambir, Borneo, including four large general flowering events

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METADATA S1

CLASS I. DATA SET DESCRIPTORS

A. Data set identity

Lambir_FFR: Long-term monitoring of plant reproductive phenology in Lambir Hills National Park

B. Data set identification code

Lambir_FFR dataset

C. Data set description

1. Originators: Shoko Sakai[1,2], Kuniyasu Momose[3†] , Teruyoshi Nagamitsu[4], Rhett D. Harrison[5], Tomoaki Ichie[6], Masahiro Nomura[7], Takakazu Yumoto[8], Hidetoshi Nagamasu[9], Runi anak Sylvester Pungga[10], Takao Itioka[11], Lucy Chong[12], Tohru Nakashizuka[4], Abang A. Hamid[13†], Tamiji Inoue[14†]

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2. Abstract: Flowering and fruiting phenology can have large impacts on a plant's reproductive success. In many plant species, these phenological events are controlled by seasonal climatic cues, resulting in one-year reproductive cycles. However, parts of SE Asian tropical forests have an aseasonal climate with irregular fluctuations. This database comprised of phenology records collected from 1993 to 2011 at the community level in an aseasonal lowland dipterocarp forest of the Lambir Hills National Park in Borneo. Observations were made every two weeks at three sites: The Canopy Biology Plot (CBP) with tree towers and walkways, the Operation Raleigh Tower (ORT) area with a tree tower for tourist attraction, and the Crane Plot located between the CBP and ORT, where plants were monitored from terraces on a canopy crane installed in 2000. The dataset includes in total 172,521 records of 450, 118, and 51 plants in CBP, Crane Plot, and ORT, respectively, representing 303 species. The number of individuals per species ranges from 1–21, and 64.9% are represented by only one. The plants in the censuses were mostly trees, but also included lianas and epiphytes. The data has been used to study the causes and consequences of synchronized flowering and fruiting at the community level, a phenomenon unique to the region. Previous studies have shown that this synchronization is synergistically driven by cool air temperature and drought. Irregular flowering and fruiting have significant impacts on flower visitors, frugivores, forest material cycling, and plant regeneration. The dataset can also be used for comparing the phenology of the same species or group among forests and regions and exploring its association with climates. One major concern regarding tropical forests in the area is the effects of climate change on this community-wide masting regime, which could disrupt forest regeneration and ecosystem processes. The dataset could be an important source of information for conservation efforts aimed at protecting these amazingly diverse forest ecosystems. This dataset can be freely used for non-commercial purposes. Users of these data should cite this data paper in any publications resulting from its use and acknowledge the Forest Department Sarawak and Sarawak Forestry Corporation.

D. Keywords/phrases

Lambir Hills National Park, Sarawak, Malaysia, Borneo, 1993–2011, Plant reproductive phenology, General flowering, Dipterocarp forest, Mast seeding

CLASS II. RESEARCH ORIGIN DESCRIPTORS

A. Overall project description

1. Identity: The data was taken as a part of "the Canopy Biology Program in Sarawak" (Roubik et al. 2005).

2. Originators: Abang A. Hamid[1†], Tamiji Inoue[2]

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2 Center for Ecological Research, Kyoto University, Hirano 2-509-3, Otsu 520-2113, Japan

Figure 1. (a) The location of the Canopy Biology Plot (CBP), Crane Plot (Crane) and Operation Raleigh Tower (ORT). The base map was provided by the courtesy of Kazuyuki Ooi. (b) A canopy walkway with a researcher on it in CBP. (c) A photograph of the forest taken from the canopy crane. (d) The 80-m tall canopy crane. It has three observation platforms at 20, 40 and 60 m above ground in addition to the one at the top (photo credits for b–d: Shoko Sakai).

3. Period of study: From 1993 to present.

4. Objectives: The objectives of the projects are: (1) to investigate the unique phenomenon of synchronized flowering fruiting at the community level in aseasonal lowland dipterocarp forests; (2) to describe interspecific interactions in the forests; and (3) to identify the effects of climatic changes on phenology and interspecific interactions.

5. Abstract: Same as Class 1.

6. Sources of funding: JSPS (Japan Society for the Promotion of Science) KAKENHI 16405006 (S Sakai), 19255006 (T Nakashizuka), 20405009 (S Sakai), 21255004 (T Itioka), 04041067 (T Inoue); the Research Institute for Humanity and Nature Project (P2–2) (T Nakashizuka); the CREST program of the Japan Science and Technology Corporation (T Nakashizuka).

B. Specific subproject description

1. Site description

The study was carried out in a lowland dipterocarp forest at Lambir Hills National Park, Borneo (4.213º N, 114.036º E, 150–250 m above sea level) (Fig. 1a, c). The annual average rainfall in the park is about 2600 mm per year (Harrison 2000). Seasonal variation in rainfall is low but noticeable (Kumagai and Kume 2012), while the area occasionally experiences droughts with biological consequences (Harrison 2000, Nakagawa et al. 2000, Itioka and Yamauti 2004). More details about the forest are available in previous literature (Sakai et al. 1999, 2006; Roubik et al. 2005).

To study plant reproductive phenology of the forest at the community level, we have been monitoring plants in three areas in the park since 1993. The first area is in and around the Canopy Biology Plot (CBP, 200 x 400 m) (Roubik et al. 2005), where plants have been observed using a canopy observation system that includes tree towers and walkways (Fig. 1a, b). The CBP includes humult and udult soils (sandy clay, light clay, or heavy clay in texture), several ridges and valleys, and closed (mature-stage) forests and canopy gaps. The second area is around another tower, which had been constructed by Operation Raleigh and donated to the park (Operation Raleigh Tower). It is situated along a stream on yellow sandstone about 1 km northeast of the CBP. The last area is the Crane Plot of 4 ha located at an intermediate spot between the above two areas, where we monitored the plants from terraces on a canopy crane installed at the center of the plot on relatively flat bedrock (Crane Plot) (Nakashizuka et al. 2003).

2. Sampling design

We initiated plant monitoring in the CBP and ORT areas in 1993, and in the Crane Plot area in 2003 (Fig. 2). Our sampling of plants was not solely based on the number of individuals of each life form, but was biased towards larger plants, such as canopy and emergent trees. Throughout the monitoring period, some plants were removed from the list due to mortality, poor health, reduced visibility caused by changes in vegetation or breakdown of the tower or walkway, or difficulty in determining their reproductive status from a distance. To maintain an adequate sample size, we occasionally added a small number of plants. Voucher specimens of the plants have been deposited at the Sarawak Herbarium, Forest Department Sarawak (Index Herbariorum: SAR), and the Kyoto University Museum (KYO).

We excluded certain plants from the dataset presented in this study, including: 1) plants of unknown family; and 2) plants for which determining reproductive status was challenging (such as *Ficus* spp., where distinguishing between flowering and fruiting stages can be difficult). As a result, the dataset includes 172,521 records of 450, 118, and 51 plants in CBP, Crane Plot, and ORT, respectively. The number of individuals per species ranged from 1–21, and 64.9% are represented by only one. Among the 619 individuals under observation, 83 did not reproduce. Some censuses may lack observations from one or two sites (Fig. 2), and the number and duration of observations varied among the plants (Fig. 3).

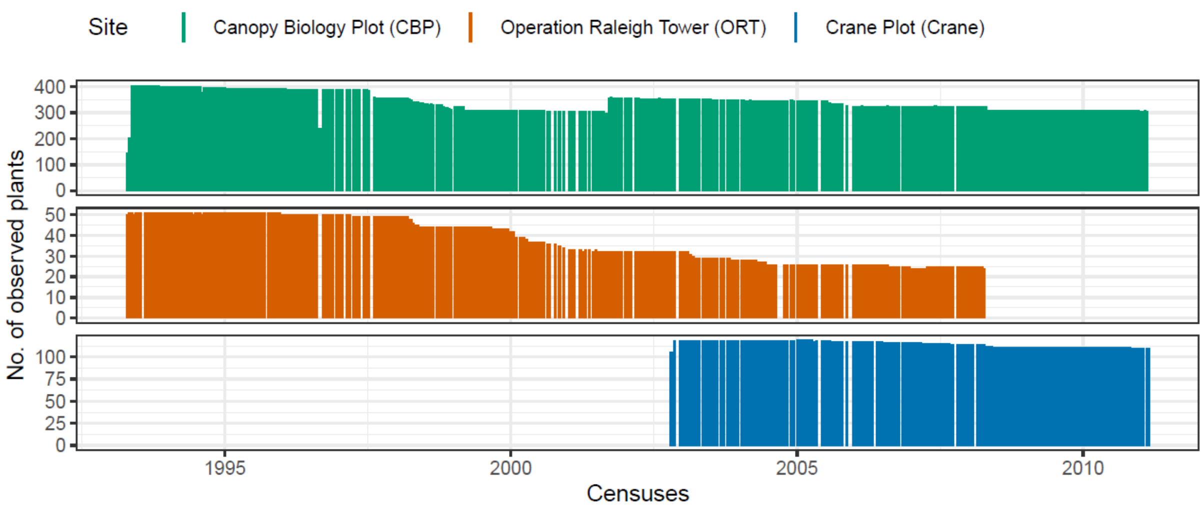

Figure 2. The changes in the number of observed plants in each census at the three sites, Canopy Biology Plot (CBP), Operation Raleigh Tower (ORT) and Crane Plot (Crane), from 1993 to 2011. We conducted 407 phenology censuses from 1993 to 2011. The number of observed plants in each census is indicated by the height of the bar. The bars are colored according to the site of the plants. Monitoring at CBP and ORT was started in 1993, and monitoring at ORT ceased in 2008. Monitoring at the Crane Plot was initiated in 2003.

To facilitate data interpretation, the plant species were classified into eight life forms (*lfm*). Most tree species were distinguished by the height of the final developmental stage of reproductive individuals: forest floor (T1, maximum height, 2.5 m), understory (T2, 2.5–12.5 m), subcanopy (T3, 12.5–27.5 m), canopy (T4, 27.5–42.5 m), and emergent (T5, 42.5 m). Forest floor plants were not included in this study. We dealt with tree species that grew only at newly made canopy gaps as gap trees (G) independently from the above five tree categories, regardless of their height. Other than trees, we distinguished epiphytes (E) and lianas (L). Among the 305 species, 13, 75, 47 and 24 species were understory, subcanopy, canopy and emergent trees, respectively. 24, 7 and 39 species were categorized as epiphytic, gap tree and liana species. We could not assign the variable for 14 species due to lack of information. The size (*size*) of the non-gap trees that were under the monitoring in 1993 (30.0 % of non-gap trees) was also recorded using the following categories: II (2.5–12.5 m), III (12.5–27.5 m), IV (27.5–42.5 m), and V (> 42.5 m). These heights were not distinguished for gap trees, epiphytes, and lianas.

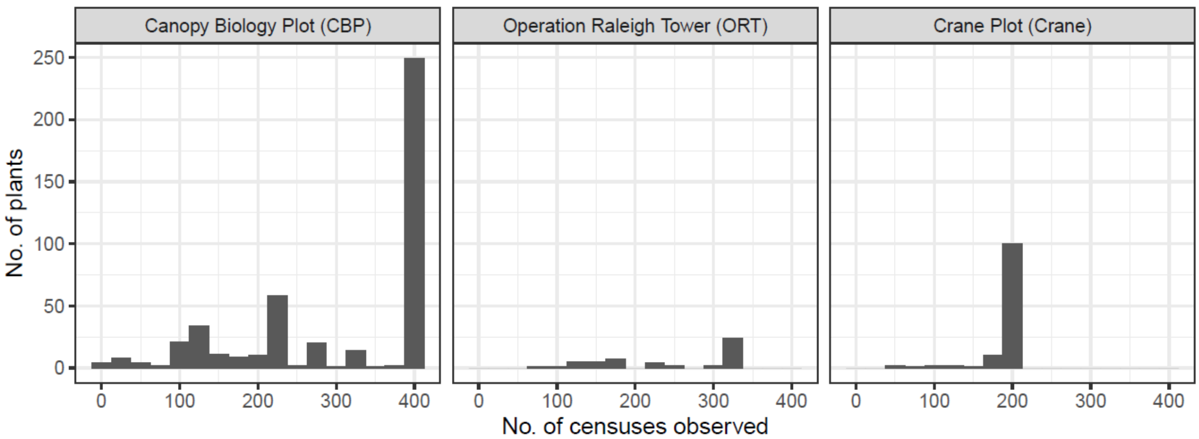

Figure 3. Histogram showing the number of censuses in which each plant was observed at the three sites. More than half of the plants at Canopy Biology Plot (CBP) were observed in almost all of the 407 censuses, while the plants at Operation Raleigh Tower (ORT) and the Crane Plot were monitored in 188 and 325 censuses at most, respectively.

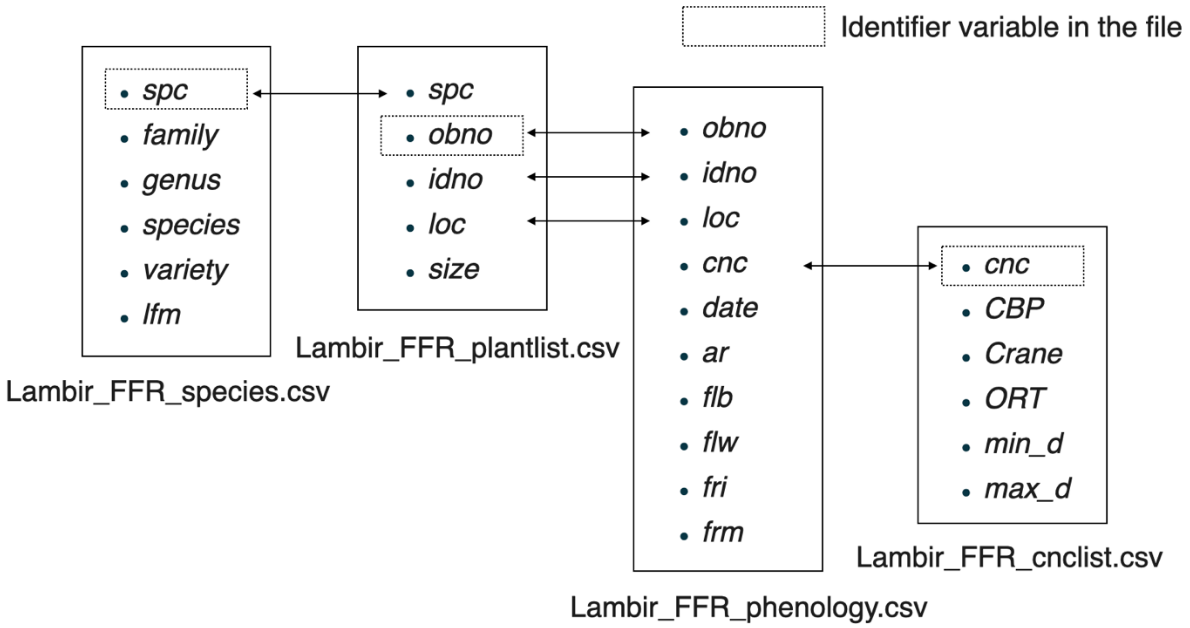

Figure 4. Relationships among the four data files and the variables they include. Variables shared between files are indicated by arrows, and identifier variables (variables without duplicates within the file) in each file are indicated by dotted rectangles. Lambir_FFR_phenology.csv, the main body of the dataset, does not have an identifier variable, while the rows have a unique combination of *obno* and *cnc*.

3. Research methods

We conducted censuses to assess the reproductive status of the plants approximately twice a month, mostly from tree towers, walkways, and canopy crane platforms (Fig. 1b, d). The censuses were conducted by the same single observer in charge at the time, while backup observers infrequently followed him to check the plants under observation and to avoid potential biases resulting from the observer's identity. On irregular and rare occasions when the principal observer took a vacation or sick leave, one of the backup researchers conducted the survey. The observer in charge was changed once between 1993 and 2011 (at the end of 1994). At that time, the new observer followed the old one several times before starting to perform observation alone.

During a census, we visited all plants under monitoring at least once. Each plant was assigned a unique plant ID (*idno*) and usually one, rarely two or three, observation IDs (*obno*). More than one observation ID of some plants is due to these plants being observed multiple times from different positions and directions in a census. Different observation IDs corresponded to different observation positions.

	obno	idno	cnc	loc	date	ar	flb	flw	fri	frm	flw_0	frt_0
108722	C0	S152	P19960124	CBP	1996-01-24	G0					0	0
108723	C0	S152	P19960205	CBP	1996-02-05	G0					0	0
108724	C0	S152	P19960219	CBP	1996-02-19	G0					0	0
108725	C0	S152	P19960307	CBP	1996-03-07	G0					0	0
108726	C0	S152	P19960326	CBP	1996-03-26	G1	P4	Pp	P0	P0	0	0
108727	C0	S152	P19960401	CBP	1996-04-01	G2	P4	Pp	P0	P0	0	0
108728	C0	S152	P19960409	CBP	1996-04-09	G2	P4	Pp	P0	P0	0	0
108729	C0	S152	P19960417	CBP	1996-04-17	G2	P2	P2	P0	P0	1	0
108730	C0	S152	P19960425	CBP	1996-04-25	G2	P0	P2	P2	P0	1	0
108731	C0	S152	P19960502	CBP	1996-05-02	G2	P0	Pp	P4	P0	0	0
108732	C0	S152	P19960508	CBP	1996-05-08	G2	P0	P0	P4	P0	0	0
108733	C0	S152	P19960521	CBP	1996-05-21	G2	P0	P0	P4	Pp	0	0
108734	C0	S152	P19960605	CBP	1996-06-05	G2	P0	P0	P4	Pp	0	0
108735	C0	S152	P19960612	CBP	1996-06-12	G2	P0	P0	P4	Pp	0	0
108736	C0	S152	P19960622	CBP	1996-06-22	G2	P0	P0	P4	Pp	0	0
108737	C0	S152	P19960704	CBP	1996-07-04	G2	P0	P0	P3	P1	0	1
108738	C0	S152	P19960714	CBP	1996-07-14	G2	P0	P0	P2	P2	0	1
108739	C0	S152	P19960725	CBP	1996-07-25	G2	P0	P0	P1	P3	0	1
108740	C0	S152	P19960808	CBP	1996-08-10	G2	P0	P0	P1	P3	0	1
108741	C0	S152	P19960830	CBP	1996-08-30	G2	P0	P0	P0	P4	0	1
108742	C0	S152	P19960919	CBP	1996-09-19	G0					0	0
108743	C0	S152	P19961007	CBP	1996-10-07	G0					0	0
108744	C0	S152	P19961021	CBP	1996-10-21	G0					0	0
108745	C0	S152	P19961107	CBP	1996-11-07	G0					0	0

Figure 5. An example of a reproductive event recorded in our dataset. The extracted raw records in the file Lambir_FFR_phenology.csv describe a series of reproductive activities from flowering to fruit dispersal of an emergent tree of *Dipterocarpus lanceolata* (Dipterocarpaceae) in 1996.

To observe a plant, the observer stood at the designated observation point and looked in a specific azimuth direction at a specific elevation angle. Then he located the plants by referring to photographs taken from the same point and with the same direction and angle, and notes on them. This procedure helped the observer to find the right plant from a canopy crane, tower or walkway at a distance. It was also critical because many observation targets did not have any tag or marking tape visible from the observation point. Then, he observed the reproductive status of the plant using binoculars and recorded it on a data sheet on which the observation IDs were printed. He also referred to the records of the preceding census in order not to overlook the development of the reproductive organs previously observed as well as to confirm previous observations. For example, leaf or flower buds difficult to distinguish at initial observations might be confirmed in a later census. When it was the first observation of flower or fruit of the plant, its photos and specimens were taken if it was possible.

The intensity and status of reproductive activity of the plants were recorded as follows: Intensity (*ar*) was described using five grades: G0 for non-reproductive, Gp for very few flowers or fruits covering less than 1% of the crown, G1 for more than 1% and less than 50%

of the crown covered with flowers or fruits, G2 for more than 50% and less than 80% of the crown covered with flowers or fruits, and G3 for flowers or fruits covering more than 80% of the crown. When reproductive organs were present, the stage of reproduction was recorded as the proportion of flower buds (*flb*), flowers (*flw*), immature fruits (*fri*), and mature fruits (*frm*) among all reproductive organs. We evaluated their proportions using five grades: P0 for absent, Pp for the proportion much less than 25%, P1 for approximately 25%, P2 for approximately 50%, P3 for approximately 75%, and P4 for nearly 100%. The grades were recorded such that the numerical sum of the integer part (0, 1, 2, 3, 4), excluding Pp, was always 4 for each plant (Fig. 5). When the individual was non-reproductive and *ar* was G0, *flb*, *flw fri* and *frm* were left blank. 13.4 % of the individuals did not reproduce during the study. In additions to these grades, the observer added comments about the reproductive status such as "fruits have been dispersed", or about the status of the plant such as "the tree may be dead".

When we described the changes in the proportion of flowering fruiting individuals, the plant was regarded as "flowering" when the grade of *ar* (all reproductive organs) was G1 or greater and the grade of *flw* (proportion of flowers) was Pp or higher. Similarly, the individual was regarded as "fruiting" when the grade of *ar* was G1 or greater and either of *fri* or *frm* (proportion of immature and mature fruits) was Pp or higher. In the case of the individuals that were observed multiple times during a census (individuals with multiple *obno* codes), the individual was regarded as flowering or fruiting when the individual was recorded to be flowering or fruiting at least once during the census (Sakai et al. 1999; Sakai et al. 2007; Fig. 6).

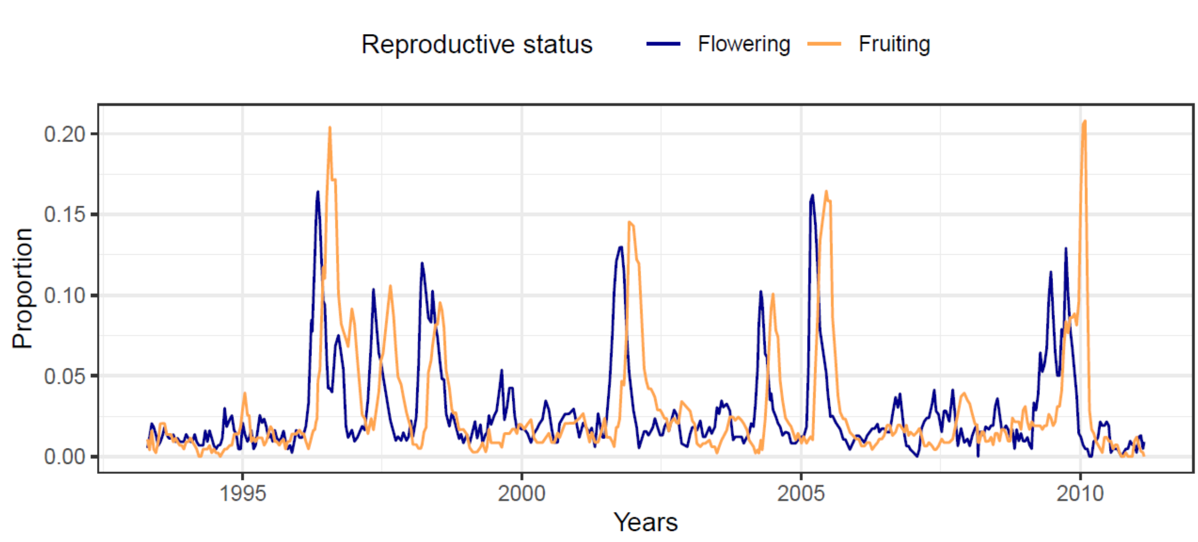

Figure 6. Changes in the proportion of the flowering and fruiting plants. A plant with reproductive organs covering 1 % or more of its canopy (the variable *ar* was G1 or greater) including at least some flowers or fruits (the variable *flw* or *frt* was either Pp, P1, P2, P3 or P4) at the census was categorized as flowering or fruiting, respectively.

CLASS III. DATA SET STATUS AND ACCESSIBILITY

A. Status

1. Latest update: November, 2023

2. Latest archive data: March, 2011. While the phenology monitoring is still ongoing, the observation methods and frequency and target plants were modified after March 2011. Therefore, the data will be published as a separate paper.

3. Metadata status: Metadata are complete.

4. Data verification: Data entry was done by the observer himself, usually within two weeks. All data was visually checked for any input errors. The data entry was also checked by SAS or R programs to find inputs outside the ranges (see Tables 1–4) and inputs that did not satisfy the rule (the total of *flb*, *flw*, *fri* and *frm* should be 4). Such unexpected entries were corrected by referring to the hand-written original datasheet. Original data sheets are kept at Kyoto University.

B. Accessibility

1. Storage location and medium: Data can be found as Supporting Information to the Ecology Data Paper and on Zenodo (Sakai et al., 2025; https://doi.org/10.5281/zenodo.13892251.

2. Contact person: Shoko Sakai, Department of Geography, Hong Kong Baptist University. Email: shokosakai@hkbu.edu.hk

3. Copyright or propriety restrictions: There are no copyright or proprietary restrictions for research or teaching proposes. Users of these data should cite this data paper in any publications resulting from its use and acknowledge the Forest Department Sarawak and Sarawak Forestry Corporation.

4. Citation: This paper.

CLASS IV. DATA STRUCTURAL DESCRIPTORS

A. Data set files

1. Identity: see the list below. The relationships between files are illustrated in Fig. 4.

- Lambir_FFR_phenology.csv
- Lambir_FFR_plantlist.csv
- Lambir_FFR_species.csv
- Lambir_FFR_cnclist.csv

2. Size

- Lambir_FFR_phenology.csv (172,521 rows + header, 10 columns, 10.5 MB)
- Lambir_FFR_plantlist.csv (633 rows + header, 5 columns, 18 KB)
- Lambir_FFR_species.csv (305 rows + header, 6 columns, 13 KB)
- Lambir_FFR_cnclist.csv (407 rows + header, 6 columns, 22 KB)

3. Format and storage: Digital data file in .csv

4. Header information: See header information in section B.

5. Alphanumeric information: Mixed.

B. Variable information

The data files have unique sets of variables. At least one variable in each file is shared with one of the other files (see Fig. 4 for the details).

Table 1. Header information of "Lambir_FFR_phenology.csv".

Variable name	Description	Nature	Range or Format
obno	Each plant has usually one, rarely two or three, observation ID (*obno*). Variation in the number of the observation IDs is because some plants were observed twice, or rarely three times, from different positions and directions. Different observation IDs of a same plant correspond to different positions of the observation.	Character	
idno	Plant ID. Each plant has a unique plant ID (*idno*).	Character	
cnc	Census ID, which is the combination of "P" and the start date of the census. All observations are assigned to one of the censuses based on the observation date.	Character	
loc	Location of the observation described by one of the three values: CBP (Canopy Biology Plot), Crane (Crane Plot), or ORT (Operation Raleigh Tower).	Categorical	CBP, Crane, ORT
date	Date of the obsevation.	Discrrete	YY-MM-DD
ar	Amount of reproductive organs. When ar is *ar*, the variables below (*flb*, *flw*, *fri* and *frm*) are blank.	Categorical	G0, Gp, G1, G2, G3
flb	Proportion of flower buds among reproductive organs.	Categorical	P0, Pp, P1, P2, P3, P4

flw	Proportion of flowers among reproductive organs.	Categorical	P0, Pp, P1, P2, P3, P4
fri	Proportion of immature fruits among reproductive organs.	Categorical	P0, Pp, P1, P2, P3, P4
frm	Proportion of mature fruits among reproductive organs.	Categorical	P0, Pp, P1, P2, P3, P4

Table 2. Header information of ″Lambir_FFR_plantlist.csv″.

Variable name	Description	Nature	Range or Format
spc	Species ID to indicate the species (and variety) identity of the plant.	Character	
obno	Observation ID. See Table 1.	Character	
idno	Plant ID. See Table 1.	Character	
loc	Location of the observation. See Table 1.	Categorical	CBP, Crane, ORT
size	Height of the tree (not aplicable to plants with the life form of E, G or L).	Categorical	II, III, VI, V

Table 3. Header information of ″Lambir_FFR_species.csv″.

Variable name	Description	Nature	Range or Format
spc	Species ID. See Table 2.	Character	
family	Family of the species.	Character	
genus	Genus name.	Character	
species	Epithet.	Character	
variety	Variety.	Character	
lfm	Life form.	Categorical	T2, T3, T4, T5, E, G, L

Table 4. Header information of "Lambir_FFR_cnclist.csv".

Variable name	Description	Nature	Range or Format
cnc	Census ID. See Table 1.	Character	
CBP	The number of observed plants in the CBP during the census.	Integer	0–405
Crane	The number of observed plants in the Crane Plot during the census.	Integer	0–119
ORT	The number of observed plants in the ORT area during the census.	Integer	0–51
min_d	First date of the census.	Date	YY-MM-DD
max_d	Last date of the census.	Date	YY-MM-DD

CLASS V. SUPPLEMENTAL DESCRIPTORS

A. Data acquisition

1. Data forms or acquisition methods: Data was recorded on a paper datasheet in the field (Fig. 1b). Then digitized at the field laboratory in the park using Microsoft Excel. Then converted into csv files.

2. Location of completed data forms: Scanned data sheets are kept at the Graduate School of Human and Environmental Studies, Kyoto University.

3. Data entry verification procedures: Procedures employed to verify that digital data set is free of errors: See Class III, A, 4.

B. Graphical/statistical data representations

C. Computer programs

Data processing was performed in the R version 4.2.3 (R Core Team 2020) using the packages "ggplot2" (Wickham 2009), "dplyr"(Wickham et al. 2022) and "tidyr" (Wickham 2021). R script for Figs 2, 3, and 6 is provided in the file names as Lambir_FFR_graph.R.

D. Lists of publications using the dataset

Hiiragi, K., Matsuo, N., Sakai, S., Kawahara, K., Ichie, T., Kenzo, T., Aurelia, D.C., Kume, T., Nakagawa, M. (2022) Water uptake patterns of tropical canopy trees in Borneo:

Species-specific and temporal variation and relationships with aboveground traits. Tree Physiology: 42, 1928–1942, https://doi.org/10.1093/treephys/tpac061

Ueno, H., Araya, K., Meleng, P., Kaling, H., Sakai, S., Kishimoto-Yamada, K., Kon, M., Itioka, T., Satake, A. (2021). Six-year population dynamics of seven passalid species in a humid tropical rainforest in Borneo. Entomological Science. https://doi.org/10.1111/ens.12490

Ushio, M., Osada, Y., Kumagai, T. O., Kume, T., Pungga, R. A. S., Nakashizuka, T., Itioka, T., Sakai, S. (2020). Dynamic and synergistic influences of air temperature and rainfall on general flowering in a Bornean lowland tropical forest. Ecological Research 35: 17-29.

Asano, I., Itioka, T., Kishimoto-Yamada, K., Shimizu-Kaya, U., Mohammad, F. B., Hossman, M. Y., Bunyok, A., Rahman, M. Y. A., Sakai, S., Meleng, P. (2017) Increased seed predation in the second fruiting event during an exceptionally long period of community-level masting in Borneo. Ecological Research 32: 537-545, https://doi:10.1007/s11284-017-1465-0.

Tokumoto, Y., Sakai, S., Matsushita, M., Ohkubo, T., Nakagawa, M. (2014). Spatiotemporal variability in the reproductive success of the continually flowering shrub *Dillenia suffruticosa* in Borneo. Biotropica 46: 583-590.

Nakagawa, M., Matsushita, M., Kurokawa, H., Samejima, H., Takeuchi, Y., Aiba, M., Katayama, A., Tokumoto, Y., Kume, T., Yoshifuji, N., Kuraji, K., Nagamasu, H., Sakai, S., Nakashizuka, T. (2012) Possible negative effect of general flowering on tree growth and aboveground biomass increment in a Bornean tropical rainforest. Biotropica 44: 715-719.

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